WASP-12b

R-1200

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_200314` · created 2026-07-31T06:31:58+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458922.67795 +/- 0.00071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.045
Transit depth (Rp/Rs)^2	2.31 +/- 1.36 [%]
Orbital Inclination (inc)	76.85 +/- 2.94
Airmass coefficient 1 (a1)	0.95 +/- 0.12
Airmass coefficient 2 (a2)	0.05 +/- 0.11
Scatter in the residuals of the lightcurve fit is	12.49 %
Best Comparison Star	#3 - [629, 114]
Optimal Aperture	2.06
Optimal Annulus	13.61
Transit Duration (day)	0.107 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.152 ± 0.045 vs 0.1178 (deviation 0.76σ)
Duration fit vs published	0.107 d vs 0.125 d (deviation 1.06σ)
Mid-time O–C	0.3 min
Depth SNR	3.4
Residual scatter	12.49 %

Light curve

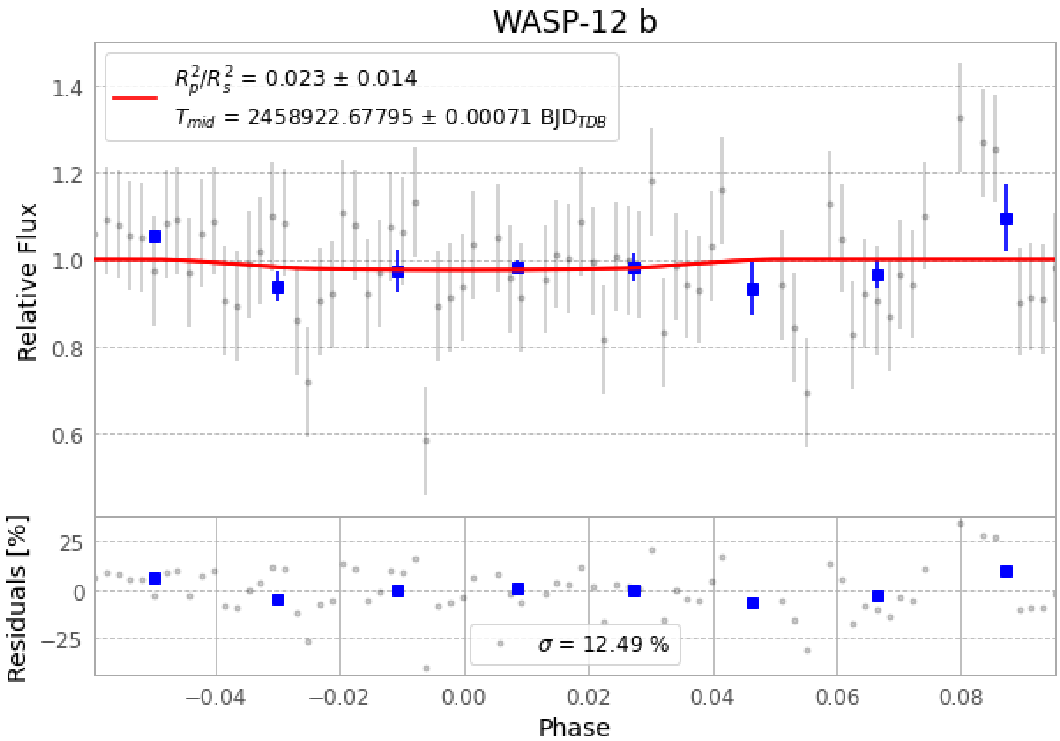

Figure 1 — FinalLightCurve_WASP-12 b_2020-03-13.png (SHA-256 b28d706279fefdef…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [439, 298], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 559ebe621773baee…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

88 FITS frames (58 MB), `WASP-12200314023912.FITS` → `WASP-12200314070012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-200314.log (57f279e23c3f…), FinalLightCurve_WASP-12 b_2020-03-13.png (b28d706279fe…), FinalLightCurve_WASP-12 b_2020-03-13.csv (8e977a9934d5…)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)